2013 - 14 iunie - A

1. Clasa e gradul de derivab.
/continuitate

$C_1, C_2$ + poveste?

2. $\langle f - g^*, g \rangle = 0 \quad \forall g \in G$.
$g^*$ unic?

$P_p \cdot g_1^* \neq g_2^*$ cele vai bune
doua prox. ale lui $f = 0$

$\langle f - g_1^*, g \rangle = \langle f - g_2^*, g \rangle = 0$
$\forall g$

$g = g_1^* - g_2^*$

$\| g_1^* - g_2^* \|^2 = \langle g_1^* - f + f - g_2^*, g \rangle$

$= \underbrace{\langle f - g_2^*, g \rangle}_{0} - \underbrace{\langle f - g_1^*, g \rangle}_{0} = 0$

$\Rightarrow g_1^* - g_2^*$

(additivitate)

3. $B_{01}(t) = (1-t)P_0 + tP_1$
$B_{12}(t) = (1-t)P_1 + tP_2$
$B_{23}(t) = (1-t)P_2 + tP_3$
$B_{02}(t) = (1-t)B_{01}(t) + tB_{12}(t)$
$B_{02}(t) = (1-t)^2 P_0 + 2(1-t)tP_1 + t^2 P_2$
$B_{13}(t) = (1-t)^2 P_1 + 2(1-t)tP_2 + t^2 P_3$

$B_{03}(t) = (1-t)B_{02}(t) + tB_{13}(t)$

$= (1-t)^3 P_0 + 2(1-t)^2 tP_1 + (1-t)t^2 P_2$
$+ (1-t)^2 tP_1 + 2(1-t)t^2 P_2 + t^3 P_3$

$B_{03}(t) = (1-t)^3 P_0 + 3(1-t)^2 tP_1$
$+ 3(1-t)t^2 P_2 + t^3 P_3, \quad t \in [0,1]$

4. $f(x_0+h) = f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 + \frac{1}{6}f'''(x_0)h^3 + \frac{1}{24}f^{(4)}(a)h^4$

$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{1}{2}f''(x_0)h^2 - \frac{1}{6}f'''(x_0)h^3 + \frac{1}{24}f^{(4)}(b)h^4$

$x_0 - h < a < x_0 < b < x_0 + h$

~~$f''(x_0)$~~

$f(x_0+h) + f(x_0-h) = 2f(x_0) + f''(x_0)h^2 + \frac{1}{24}f^{(4)}(a)h^4 + \frac{1}{24}f^{(4)}(b)h^4$

$f''(x_0) = \frac{1}{h^2}\left[f(x_0-h) - 2f(x_0) + f(x_0+h)\right] - \frac{h^2}{24}\left[f^{(4)}(a) + f^{(4)}(b)\right]$

$\exists c \in (a,b)$ a.î. $f^{(4)}(c) = \frac{1}{2}\left[f^{(4)}(a) + f^{(4)}(b)\right]$

$\Rightarrow f''(x_0) = \frac{1}{h^2}(\dots) - \frac{h^2}{12}f^{(4)}(c)$

formulă de derivare de două ori în punct de mijloc

5. ✓
6. ✓
7. ✓
8. Nu se face

2016 – (Hiut Bous Nu Neville)

Extrap. R.

1. $M = N_1(h) + k_1h + k_2h^2 + \dots$ valabilă pt orice $h > 0$

$\Rightarrow h/2$ formulă $O(h)$

$M = N_1\left(\frac{h}{2}\right) + k_1\frac{h}{2} + k_2\frac{h^2}{4} + \dots$

$M = N_1\left(\frac{h}{2}\right) + \left[N_1\left(\frac{h}{2}\right) - N_1(h)\right] + k_2\left(\frac{h^2}{2} - h^2\right) + k_3\left(\frac{h^3}{4} - h^3\right) + \dots$